

Elementary bounds on digital sums of powers

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March 28, 2026

Some notation

Elementary
bounds on
digital sums of
powers

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Sierpinski's
problem

Generalization
to Other
Bases

Stewart's
Theorem

For integers $N \geq 0$ and $b \geq 2$, the *base- b expansion* of N is the unique representation of the form

$$N = \sum_{i=0}^{\infty} d_i b^i, \quad d_i \in \{0, 1, \dots, b-1\}.$$

The integers d_i are the *base- b digits* of N ; all but finitely many of these digits are zero.

We write $s_b(N)$ for the sum of the base- b digits of N , and $c_b(N)$ for the number of nonzero digits in that expansion.

A question to start with

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digital sums of
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to Other
Bases

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Does the sum of decimal digits of 2^n increase to infinity with n ?

n	2^n	$s_{10}(2^n)$
5	32	5
10	1024	7
15	32768	26
20	1048576	31
25	33554432	29

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It looks like $\lim_{n \rightarrow \infty} s_{10}(2^n) = \infty$, but the sequence is not monotonic.

The last digit of 2^n

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digital sums of
powers

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Sierpinski's
problem

Generalization
to Other
Bases

Stewart's
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We write the decimal expansion of 2^n as

$$2^n = \sum_{i=0}^{\infty} d_i 10^i,$$

where each digit satisfies $d_i \in \{0, 1, \dots, 9\}$.

First easy fact

The last digit d_0 is nonzero, because a power of 2 is never divisible by 10.

So every 2^n already has *at least one* nonzero digit.

The first block of digits

Assume $n \geq 4$. Then 2^n is divisible by $2^4 = 16$.

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digital sums of
powers

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Sierpinski's
problem

Generalization
to Other
Bases

Stewart's
Theorem

The first block of digits

Assume $n \geq 4$. Then 2^n is divisible by $2^4 = 16$.

Look at the last four digits of 2^n :

$$2^n \bmod 10^4.$$

This remainder is divisible by 16.

Elementary
bounds on
digital sums of
powers

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Sierpinski's
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Generalization
to Other
Bases

Stewart's
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Elementary
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digital sums of
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Sierpinski's
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Generalization
to Other
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Now suppose the three digits d_1, d_2, d_3 were all zero. Then the last four digits would look like

$$000d_0,$$

so the remainder would be less than 10, which is impossible.

The first block of digits

Elementary
bounds on
digital sums of
powers

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Sierpinski's
problem

Generalization
to Other
Bases

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Conclusion

At least one of the digits d_1, d_2, d_3 must be nonzero.

The second block of digits

Now assume $n \geq 14$. Then 2^n is divisible by 2^{14} , and

$$2^{14} = 16384 > 10^4.$$

Look at the last 14 digits:

$$2^n \bmod 10^{14}.$$

This remainder is divisible by 2^{14} .

The second block of digits

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$$2^{14} = 16384 > 10^4.$$

Look at the last 14 digits:

$$2^n \bmod 10^{14}.$$

This remainder is divisible by 2^{14} .

Now suppose the ten digits $d_4, \dots, d_{13}$ were all zero. Then the last 14 digits would look like

$$0000000000d_3d_2d_1d_0$$

so the remainder would be less than 10^4 , which is a contradiction.

The second block of digits

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Conclusion

At least one digit in the block $d_4, d_5, \dots, d_{13}$ is nonzero.

What is the pattern?

We keep choosing larger and larger cutoffs:

$$4, 14, 47, 157, \dots$$

so that each new power of 2 is bigger than the previous power of 10.
For example,

$$2^{14} > 10^4, \quad 2^{47} > 10^{14}, \quad 2^{157} > 10^{47}.$$

Then the same argument says:

$$\begin{aligned} n \geq 4 &\implies d_1 + d_2 + d_3 > 0, \\ n \geq 14 &\implies d_4 + \dots + d_{13} > 0, \\ n \geq 47 &\implies d_{14} + \dots + d_{46} > 0, \text{ etc.} \end{aligned}$$

Why this works

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Sierpinski's
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Generalization
to Other
Bases

Stewart's
Theorem

As n grows, more and more of these disjoint blocks become available.

Each block contains *at least one* nonzero digit.

So the number of nonzero digits of 2^n tends to infinity.

Consequently, the sum of digits of 2^n also tends to infinity.

Generalization to other bases

Since we are using no special properties of 2 and 10 (except $2 \mid 10$) we can generalize this result beyond $a = 2$ and $b = 10$.

Theorem (Radcliffe)

Let $2 \leq a < b$ with $a \mid b$. Suppose N is divisible by a^n but not b . Then there exists $C > 0$ (depending on a, b) such that

$$s_b(N) \geq c_b(N) > C \log n$$

for n sufficiently large.

General Bases

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Sierpinski's
problem

Generalization
to Other
Bases

Stewart's
Theorem

The condition that $a \mid b$ can be weakened considerably.

Corollary

Let $a, b \geq 2$. Let d be the smallest factor of a with $\gcd(a/d, b) = 1$. If $\log(d)/\log(b)$ is irrational, then

$$s_b(a^n) \geq c_b(a^n) > C \log n$$

for sufficiently large n .

Example: $a = 6$ and $b = 12$.

Stewart's Theorem

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bounds on
digital sums of
powers

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Sierpinski's
problem

Generalization
to Other
Bases

Stewart's
Theorem

Theorem (Stewart, 1980)

Let $a, b \geq 2$ with $\log(a)/\log(b)$ irrational. Then

$$s_b(a^n) \geq c_b(a^n) > \frac{\log n}{\log \log n + C} - 1$$

for all $n > 4$, where C depends only on a and b .

Stewart's Theorem

Elementary
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digital sums of
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Sierpinski's
problem

Generalization
to Other
Bases

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Let $a, b \geq 2$ with $\log(a)/\log(b)$ irrational. Then

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for all $n > 4$, where C depends only on a and b .

Note: The irrationality condition is necessary!

If $\log(a)/\log(b) = r/s \in \mathbb{Q}$, then

$$a^{ns} = b^{nr}$$

has only one nonzero digit in base b .

Comparison with our results

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bounds on
digital sums of
powers

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Sierpinski's
problem

Generalization
to Other
Bases

Stewart's
Theorem

- The condition of Stewart's theorem more general than ours.
- However, the conclusion only gives a sublogarithmic lower bound.
- Our results require only elementary number theory.
- Stewart's theorem requires advanced methods.

Baker's Theorem

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Sierpinski's
problem

Generalization
to Other
Bases

Stewart's
Theorem

The proof of Stewart's theorem requires a deep result from transcendental number theory (linear forms in logarithms):

Theorem (Baker-Wüstholz, 1993)

Let $\Lambda = b_1 \log \alpha_1 + \cdots + b_n \log \alpha_n$ where:

- $b_1, \dots, b_n$ are integers with $|b_i| \leq B$
- $\alpha_1, \dots, \alpha_n$ are algebraic with heights $\leq A_i$

If $\Lambda \neq 0$, then

$$\log |\Lambda| > -(16nd)^{2(n+2)} \log A_1 \cdots \log A_n \log B$$

where $d = [\mathbb{Q}(\alpha_1, \dots, \alpha_n) : \mathbb{Q}]$.

Proof Strategy

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bounds on
digital sums of
powers

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Sierpinski's
problem

Generalization
to Other
Bases

Stewart's
Theorem

Let

$$a^n = b^m (d_1 b^{-m_1} + \cdots + d_k b^{-m_k}),$$

where $m = \lceil \log_b a^n \rceil$, $k = c_b(a^n)$, $d_i \in \{1, \dots, b-1\}$, and

$$1 = m_1 < \cdots < m_k \leq m.$$

Truncate the base- b expansion of a^n at the i -th nonzero digit:

$$a^n = b^{m-m_i} q_i + r_i$$

Proof Strategy

Elementary
bounds on
digital sums of
powers

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Sierpinski's
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Generalization
to Other
Bases

Stewart's
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Truncate the base- b expansion of a^n at the i -th nonzero digit:

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Goal: Show digit gaps m_{i+1}/m_i cannot be too large

Main steps of proof

Elementary
bounds on
digital sums of
powers

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Sierpinski's
problem

Generalization
to Other
Bases

Stewart's
Theorem

- 1 Approximate a^n by truncating to the i -th nonzero digit.

Main steps of proof

Elementary
bounds on
digital sums of
powers

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Sierpinski's
problem

Generalization
to Other
Bases

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Theorem

- 1 Approximate a^n by truncating to the i -th nonzero digit.
- 2 Estimate the digit gap m_{i+1}/m_i in terms of the truncation error $a^{-n}r_i$.

Main steps of proof

Elementary
bounds on
digital sums of
powers

David G.
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Sierpinski's
problem

Generalization
to Other
Bases

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Theorem

- 1 Approximate a^n by truncating to the i -th nonzero digit.
- 2 Estimate the digit gap m_{i+1}/m_i in terms of the truncation error $a^{-n}r_i$.
- 3 Apply Baker's theorem to obtain a lower bound on the truncation error.

Main steps of proof

Elementary
bounds on
digital sums of
powers

David G.
Radcliffe

Sierpinski's
problem

Generalization
to Other
Bases

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- 2 Estimate the digit gap m_{i+1}/m_i in terms of the truncation error $a^{-n}r_i$.
- 3 Apply Baker's theorem to obtain a lower bound on the truncation error.
- 4 Deduce thereby an upper bound on the digit gap.

Main steps of proof

Elementary
bounds on
digital sums of
powers

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Sierpinski's
problem

Generalization
to Other
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- 1 Approximate a^n by truncating to the i -th nonzero digit.
- 2 Estimate the digit gap m_{i+1}/m_i in terms of the truncation error $a^{-n}r_i$.
- 3 Apply Baker's theorem to obtain a lower bound on the truncation error.
- 4 Deduce thereby an upper bound on the digit gap.
- 5 Compute a lower bound on the number of gaps, and hence the number of nonzero digits.